SURYA S

ARTIFICIAL INTELLIGENCE ENGINEER, B.E. COMPUTER SCIENCE & ENGINEERING

Coimbatore / Mannargudi, Tamil Nadu, India, Email: surya25suresh2006@gmail.com, LinkedIn: linkedin.com/in/surya-suresh
GitHub: github.com/Surya252006, Portfolio: surya-s-portfolio-e8625.web.app

EXECUTIVE SUMMARY

Motivated early-career **Artificial Intelligence Engineer** and Computer Science undergraduate (B.E. 2024 – 2028) with demonstrated experience in applied Machine Learning, IoT sensor telemetry ingestion, and scalable full-stack web architectures. Hands-on in designing healthcare diagnostic triage systems (**MedPulse AI**), real-time agricultural telemetry classification achieving **91.4% accuracy**, and hackathon award solutions (**SMARTATHON'26 AI Scheme & Scholarship Matcher**). Proficient in Python, PyTorch, TensorFlow, Scikit-Learn, Java, React, FastAPI, Node.js, and MongoDB Atlas. IIT Guwahati AI & ML Certification qualified, committed to engineering reliable, verified, and human-centric software.

TECHNICAL SKILLS

AI & Machine Learning:	Python, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Data Preprocessing, Feature Engineering, Supervised Learning, NLP Intent Classification, Model Metrics Evaluation, Predictive Analytics.
Programming Languages:	Python, Java (Core OOP, Collections, Multithreading), C, C++, JavaScript (ES6+), TypeScript, SQL.
Web & Full-Stack Development:	React 18, Vite, Node.js, Express.js, REST APIs, FastAPI (Asynchronous Endpoints), HTML5, CSS3, Tailwind CSS, Bootstrap, EJS.
Databases & Cloud Tools:	MongoDB Atlas (Atomic Operators, Aggregations), Firebase (Firestore, Hosting, Authentication), SQLite, Git, GitHub, Postman.
Hardware & IoT Telemetry:	ESP32 Microcontrollers, Arduino, Physical Sensor Ingestion (NPK, Soil Moisture, pH), Serial Telemetry Streaming, REST Webhooks.
Core CS Foundations:	Data Structures & Algorithms, Object-Oriented System Architecture, Clean Code Principles, Linux CLI, Agile Development.

FEATURED ARTIFICIAL INTELLIGENCE & MACHINE LEARNING PROJECTS

MedPulse AI — Healthcare Symptom Triage System GitHub Repository, 2025
Technologies: Python, PyTorch, TensorFlow, Scikit-Learn, FastAPI, REST APIs

- Architected an applied clinical symptom classification engine analyzing multi-symptom input vectors to categorize patient urgency levels and risk priorities.
- Implemented lightweight, asynchronous RESTful inference microservices with FastAPI, achieving latency under 150ms for clinical triage scoring and risk categorization.
- Built structured error boundaries and standardized JSON health metric outputs to facilitate seamless interoperability with hospital triage portals.

SMARTATHON'26 — AI Scheme & Scholarship Eligibility Matcher DSU × ONEYES Hackathon, 2026
Technologies: FastAPI, Python 3.13, React 18, MongoDB Atlas, Tailwind CSS

- Engineered an intelligent scheme-matching engine for SMARTATHON'26 (Problem Statement #6), evaluating student academic, income, and demographic criteria.
- Designed deterministic rule-matching algorithms against central and Tamil Nadu state scholarship catalogs with an automated 96% qualification scoring rubric.
- Developed an interactive React 18 UI with real-time eligibility progress trackers, dynamic document checklists, and application deadline alerts.

Smart Precision Soil Telemetry & Crop Recommendation ML IoT & ML Telemetry, 2025
Technologies: Python, PyTorch, Scikit-Learn, IoT ESP32, FastAPI

- Built an end-to-end telemetry pipeline streaming real-time physical soil sensor data (nitrogen, phosphorus, potassium, moisture, and pH) via ESP32 microcontrollers.
- Trained multi-parameter classification models achieving 91.4% crop variety compatibility accuracy and precise bio-fertilizer dosage recommendations.
- Mitigated chemical runoff through algorithmic soil health monitoring, visualizing historical fertility trends via responsive web dashboards.

Personal Finance AI Assistant & Conversational Bot GitHub Repository, 2024
Technologies: Python, NLP Intent Parsing, Scikit-Learn, React 18

- Developed a conversational personal finance assistant utilizing natural language query parsing to categorize spending and suggest budget optimization rules.
- Engineered interactive visual distribution summaries and dynamic savings goal trackers with responsive chat interfaces.

FULL-STACK & SYSTEMS ENGINEERING PROJECTS

KisanConnect — AgriTech Direct Marketplace Platform kisanconnect-2267e.web.app, 2024 – Present

Technologies: React 18, Node.js, Express.js, MongoDB Atlas, Firebase Hosting

- Deployed live to production on Firebase Hosting (kisanconnect-2267e.web.app), directly bridging farmers with bulk and retail buyers to eliminate middleman markups.
- Implemented real-time APMC Mandi price comparison indices, empowering growers to benchmark market trends before listing agricultural produce.
- Engineered race-condition-free inventory decrementing utilizing MongoDB atomic operators ($gte, $inc) across multi-role access tiers (Farmer, Retailer, Admin).

Console Banking & Atomic Ledger Engine GitHub Repository, 2024

Technologies: Java, OOP Design Patterns, File Persistence, Input Validation

- Developed an enterprise console banking engine demonstrating strict Object-Oriented Principles (encapsulation, abstraction, inheritance, polymorphism).
- Built an isolated double-entry transaction journal with atomic credit/debit validation, balance guarantees, and permanent file persistence.

PosterKadai — Production E-Commerce Storefront posterkadai-e364b.web.app, 2025 – Present

Technologies: React 18, Firebase Firestore, WhatsApp API, Tailwind CSS

- Shipped production commercial storefront (posterkadai-e364b.web.app) featuring responsive catalog exploration, instant cart state, and direct WhatsApp checkout integration.
- Optimized frontend assets to sustain sub-1.2s page load speeds with continuous client engagement across regional demographics.

PROFESSIONAL EXPERIENCE & INTERNSHIPS

Full-Stack Development Intern, Cognifyz Technologies Remote, 2024

- Engineered responsive client-side user interfaces and integrated backend RESTful endpoints with Node.js, Express, and EJS following industry best practices.
- Implemented interactive form validations, asynchronous data fetching pipelines, and modular component architectures across 5 production task deliverables.

Founder & Lead Technology Developer, PosterKadai (Commercial E-Commerce) Tamil Nadu, India, 2025 – Present

- Architected the entire digital product infrastructure, cloud hosting (Firebase), database rules (Firestore), and customer WhatsApp order fulfillment pipeline.
- Handled end-to-end technical execution from system architecture to production deployment, managing digital reach and automated catalog pipelines.

Program Organizer & Technical Curriculum Lead, Image Color Lab Online, 2025 – 2026

- Structured and delivered an intensive 15-day online technical and design internship program for creative developers and designers.
- Designed training evaluation frameworks, managed participant operations and certification, and coordinated hands-on project workflows.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering 2024 – 2028

Anna University Affiliated Institution, Coimbatore, Tamil Nadu, India

- Core Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java/C++), Artificial Intelligence Fundamentals, Database Systems (RDBMS & NoSQL), Computer Networks, Operating Systems.
- Active hands-on software development alongside academic curriculum; building and deploying production-grade systems since 1st year.

HONORS, CERTIFICATIONS & ACHIEVEMENTS

- **IIT Guwahati AI & ML Certification Qualified:** Advanced qualification in applied machine learning, statistical modeling, and neural computing foundations.
- **SMARTATHON'26 Hackathon Finalist:** DSU × ONEYES Hackathon (Problem Statement #6) — Engineered full-stack AI Scholarship & Scheme Matcher.
- **Full-Stack Engineering Certification:** Completed comprehensive development internship track at Cognifyz Technologies.
- **Active Open-Source Builder:** 450+ verified Git commits and 9 publicly accessible technical repositories on GitHub (github.com/Surya252006).
- **Production Deployments:** Deployed multiple live applications running 24/7 on Google Cloud / Firebase Hosting with real active users.